MIL-A-28768/1B(EC)
30 March 1984
SUPERSEDING
MIL-A-28768/1A(EC)
2 February 1973

MILITARY SPECIFICATION

ANTENNA, FIXED, HIGH FREQUENCY, VERTICAL OMNIDIRECTIONAL BROADBAND

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the antenna described herein shall consist of this specification and the latest issue of MIL-A-28768.

1. SCOPE

1.1 Scope. This specification covers requirements for a fixed, high frequency (HF) vertical omnidirectional broadband antenna, hereinafter referred to as the antenna. The antenna is for use at shore stations for communication purposes in the HF region of the radio spectrum.

1.2 Classification. Antennas shall be of the following types, as specified (see 6.2):

TYPE	Frequency range
I	2.0 Megahertz (MHz) to 30.0 MHz
II	2.5 MHz to 30.0 MHz
III	3.0 MHz to 30.0 MHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

W-R-550	Rod, Ground (With Attachments)
QQ-C-576	Copper Flat Products With Slit, Slit And Edge-rolled, Sheared, Sawed, Or Machined Edges, (Plate, Bar, Sheet, And Strip)

MILITARY

MIL-A-28768	Antennas, Fixed, High Frequency, General Specification For

(Copies of specifications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5985

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3. REQUIREMENTS

3.1 General. The antenna shall comply with MIL-A-28768 and the requirements specified herein. The antenna shall be furnished as a complete system ready for installation and operation. The antenna shall be designed for broadband operation without tuning or adjustment.

3.2 Size. The antenna shall comply with the size limitations specified in TABLE I.

TABLE I. Size of antennas.

	TYPE I m 1/ (ft) 2/	TYPE II m (ft)	TYPE III m (ft)
Ground screen radius	One-fourth of lowest wavelength	One-fourth of lowest wavelength	One-fourth of lowest wavelength
Maximum length, width, or diameter, including guys and anchors	137 (450)	119 (390)	91 (300)
Maximum height	38 (125)	32 (105)	27 (90)

1/ Meters
2/ Feet

3.3 Environmental conditions. The antenna shall conform to the environmental requirements specified in MIL-A-28768. The antenna shall be designed so that snow and ice will not accumulate in the apex to the extent that allowable stress requirements specified in the Structural requirements paragraph of MIL-A-28768 are exceeded or the electrical requirements are degraded, or both.

3.4 Structural requirements. Structural requirements shall be in accordance with 3.4.1 through 3.4.5.

3.4.1 Rotational stability. The antenna shall be designed with an appropriate guying or bracing system which shall eliminate the possibility of rotational instability. The antenna shall be designed so that rotation about a vertical axis through the center of the antenna shall be not greater than 30 minutes of arc.

3.4.2 Field adjustment of cable tension. The design of the antenna shall provide facilities for field adjustment of cable tensions both during and after erection.

3.4.3 Antenna installation characteristics. The antenna shall be designed so that installation grading will not be required, unless within the area of the ground screen:

a. The ground roughness exceeds ±0.9144 m (±3 ft)
b. The overall ground slope exceeds ±3 degrees

Support towers shall remain vertical.

3.4.4 Wooden poles. Wooden poles shall not be used or required.

3.4.5 Materials. Materials, such as support towers and guy assemblies, and additional anchors, attachment hardware, and so forth, necessary to permit installation over a site ranging from flat and level to the limits specified in 3.4.3, shall be supplied.

3.5 Electrical performance. The antenna shall comply with the electrical design requirements of MIL-A-28768 and the performance requirements specified in 3.5.1 through 3.5.6. The antenna shall be assumed to be over perfectly conducting earth.

3.5.1 Power. Unless otherwise specified (see 6.2), Power 50 as specified in MIL-A-28768 shall apply.

3.5.2 Frequency range. The antenna shall be capable of operating without adjustment over the entire frequency range, specified by TYPE, in 1.2.

3.5.3 Gain. Gain shall be as specified in 3.5.3.1 and 3.5.3.2.

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3.5.3.1 Elevation gain. The antenna gain at the elevation beam maximum shall be not less then +4 decibels above an isotropic (dBi) radiator in free space. The gain between zero degrees elevation (that is, at the horizon) and +15 degrees elevation shall be not less than +2 dBi.

3.5.3.2 Azimuth gain. Azimuth gain shall not vary more than ±1 decibel (dB) at a given elevation angle between zero degrees and 20 degrees; that is, the antenna shall be omnidirectional in azimuth gain within ±1 dB.

3.5.4 Polarization. The antenna shall be vertically polarized.

3.5.5 Efficiency. When installed over imperfectly conducting earth, the efficiency of the antenna as specified in the Definitions paragraph of MIL-A-28768 shall be not less than 90 percent (72 pecent when the effects of voltage standing wave ratio are considered).

3.5.6 Base insulators. A minimum of six insulators (in parallel) shall be used at the apex of the cone. The insulators shall be designed so that the electrical and structural stresses are equally distributed.

3.6 Ground system. A ground system in accordance with MIL-A-28768 and 3.6.1 shall be supplied as part of the antenna.

3.6.1 Construction. The ground system shall consist of radials extending from the center (vertical) axis of the antenna. The length of each radial shall be as specified in TABLE 1. The wires shall be equally spaced at a minimum of 3-degree intervals. A minimum of 120 radials per antenna shall be required. At the outer extremity, every third radial shall be terminated to a ground rod, conforming to Type II, Class B of W-R-550, except that the diameter shall be 15.875 millimeters (mm) (0.625 inch (in.)) and the length shall be 1.829 m (6 ft). A circumferential wire of the same gage and characteristics as the radials shall be supplied for bonding to the ends of the unterminated radials and to the ground rods. The inner termination of each radial shall be to a copper plate conforming to QQ-C-576. The plate may be either circular or rectangular, shall be a minimum of 2.13 mm by 0.0164 centimeters (cm) (7 square ft by 0.0647 in.) thick, and shall cover the top and overlap the sides of a concrete foundation. The radials shall maintain separation up to their point of attachment to the plate. The inner and outer attachments of the radial wires shall be designed to be made in the field utilizing brazing. The inner attachment shall be above ground level and shall be holes in, or eyes on, the plate to attach the wires so that the solder is not the sole means of bonding the wires to the plate. The radials need not be precut or preassembled. All materials necessary to make the connections shall be supplied. For distances less than 7.62 m (25 ft) from the center of the antenna, the radials shall be designed for placement on the earth's surface. For distances greater than 7.62 m (25 ft) from the center of the antenna, the radials shall be buried to a maximum of 15.24 cm (6 in.).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. Quality assurance provisions shall be in accordance with MIL-A-28768 and as specified in 4.2.

4.2 First article inspection. In addition to the first article inspections specified in MIL-A-28768, the inspections in a through c shall be conducted:

a. Verification of size requirements of 3.2
b. Verification of rotational stability requirement of 3.4.1
c. Verification of compliance to the installation grading requirements of 3.4.3

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preparation of delivery shall be as specified in MIL-A-28768.

6. NOTES

6.1 Intended use. The antenna will be used for general purpose radio communications and is intended for permanent installation at shore locations.

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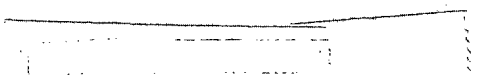
6.2 Ordering data. In addition to the items specified in MIL-A-28768, procurement documents should specify:

a. Type (see 1.2)
b. Power, if other than Power 50 specified in MIL-A-28768 (see 3.5.1)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy-EC

(Project 5985-N503-1)



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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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